

Class 10 · Practice Test (Maths and Science)

Time: 1 hour · Maximum marks: 25 · Answer all questions. Show your working.

Q1.	A cross made between two pea plants produces 50% tall and 50% short pea plants. The gene combination of the parental pea plants must be: (A) Tt and Tt (B) TT and Tt (C) Tt and tt (D) TT and tt	[1]
Q2.	Assertion (A): If zeroes of the polynomial $(2k - 1)x^2 + 4x - 3$ are reciprocal of each other, then $k = -1$. Reason (R): If $a = c$, then zeroes of the polynomial $ax^2 + bx + c$, $a \neq 0$ are reciprocal of each other. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.	[1]
Q3.	Translate the following statements into chemical equations and then balance them: (a) Nitric acid reacts with calcium hydroxide to form calcium nitrate and water. [1] (b) Sodium chloride reacts with silver nitrate to form silver chloride and sodium nitrate. [1]	[2]
Q4.	If $\tan \theta = 1/\sqrt{5}$, then find the value of $(\operatorname{cosec}^2\theta - \sec^2\theta)/(\operatorname{cosec}^2\theta + \sec^2\theta)$.	[2]
Q5.	A. State the Right-Hand Thumb Rule and describe how it helps determine the direction of magnetic field around a current-carrying conductor. [2] B. Explain how the magnetic field changes with distance from the conductor. [1]	[3]
Q6.	Find the sum of integers between 1 and 400 that are multiples of 4 as well as of 5.	[3]
Q7.	What changes are observed in the uterus subsequent to implantation of the young embryo in a human female?	[2]
Q8.	A tent is in the shape of a cylinder surmounted by a conical top. If the height and radius of the cylindrical part are 3 m and 14 m respectively, and the total height of the tent is 13.5 m, then find the area of the canvas required for making the tent, keeping a provision of 26 m^2 of canvas for stitching and wastage. Also, find the cost of the canvas to be purchased at the rate of ₹ 250 per m^2 . (Take $\pi = 22/7$.)	[5]

- Q9. Case study: In a technology park in Hyderabad, a company installed a circular meditation garden for its employees. Arjun, a designer, stood at a point P outside the garden and drew two tangents PA and PB to the circular boundary. The radius of the garden is 9 m, and the distance of point P from the centre O is 18 m. For preparing safety guidelines, Arjun needed the lengths of the tangents. During a demonstration to interns, he asked them to compute the angle formed between the two tangents using the properties of circle. [4]
- Based on the above information, answer the following questions:
- (i) What is the measure of $\angle OAP$? [1]
 - (ii) Calculate the length of the tangent PA. [1]
 - (iii) Calculate the measure of $\angle AOB$. [2]
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- Q10. Consider two lamps A and B of rating 50 W; 220 V and 25 W; 220 V respectively. Find the ratio of the resistances of the two lamps (i.e. $R_A : R_B$). [2]
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